

The Economics of Science in the US

What are the relationships between these entities?

(Funding, Advises, Reporting, Programming, Appoints/Directs, Policy, Lobbies)

The Profession

Private Foundations

National
Academy of
Science,
Engineering
and Medicine
(NASEM)

Congress

Federal Agencies
NASA, DOE, NSF

The Taxpayer

The Profession: Astro2020 Decadal N.2.1

The community of scientists, engineers, technicians, and nontechnical people engaged in the production and instruction of astronomical knowledge, as well as learners on the path to joining their ranks.

The National Academy of Sciences

The [Act to Incorporate the National Academy of Sciences](#), also known as the Academy Charter, was signed by President Lincoln on 3 March 1863, establishing the Academy as an **independent** honorary and **consulting** body with its own governance and structure.

The Act also sets out an **obligation** for the Academy to provide **scientific and technical advice to any department of the Government**, whenever requested, with the Academy receiving **no compensation for its services**.

<https://www.nasonline.org/about-the-nas/leadership/governing-documents/>

National Academy of Sciences, Engineering and Medicine (NASEM)

- The National Research Council was initially established by [Executive Order](#) of President Wilson as a contingency organization charged with **coordinating American scientific and technological research** in the service of **preparedness for war**. After the war in question—the First World War—ended in 1918, the Research Council was given permanent status as the Academy's operating arm.
- Now: The National Research Council (NRC) is operated by the National Academies of Sciences, Engineering, and Medicine.
- NASEM → NAS, NAE, NAM → NRC being the research arm of it all.

National Academy of Sciences, Engineering and Medicine (NASEM)

1916 Executive Order:

- #2 To survey the **larger possibilities of science**, to formulate comprehensive projects of research, and to develop effective means of utilizing the scientific and technical resources of the country for dealing with these projects.
- #3 To **promote cooperation** in research, at home and abroad, in order to secure concentration of effort, **minimize duplication**, and stimulate progress; but in all cooperative undertakings to give encouragement to individual initiative, as fundamentally important to the advancement of science.
- #4 To serve as a means of bringing American and foreign investigators **into active cooperation with the scientific and technical services of the War and Navy Departments** and with those of the civil branches of the Government

National Academy of Sciences, Engineering and Medicine (NASEM)/ National Research Council

- [NAS Strategic Plan](#)

- Example: Covid-19 Pandemic

“During the COVID-19 pandemic, the NRC rose to the challenge of providing peer reviewed, expert advice on time scales of a few weeks or less in response to targeted questions from health authorities.”

Example: Oversee the Creation of Consensus Reports

- Astro 2020 Decadal
- Planetary Sciences 2023 Decadal

National Academy of Sciences, Engineering and Medicine (NASEM)

Membership by nomination.

Members at UA: Renu Malhotra, Marcia Rieke, George Rieke, Roger Angel

Committee on Astronomy and Astrophysics (CAA)

- <https://www.nationalacademies.org/projects/DEPS-SSB-16-09>
- Consensus/Advisory Committee to NASEM
- The Committee on Astronomy and Astrophysics (CAA) provides an independent, authoritative forum for identifying and discussing issues in astronomy and astrophysics with the research community, the federal government, and the interested public. The committee supports scientific progress in astronomy and astrophysics by providing advice to the federal government on the implementation of [decadal survey recommendations](#).
- Rob Kennicutt is on this committee (Marcia was before)

Committee on Astrobiology and Planetary Science (CAPS)

- <https://www.nationalacademies.org/projects/DEPS-SSB-16-08/resources>
- CAPS provides an independent, authoritative forum for identifying and discussing issues in astrobiology and planetary science with the research community, the federal government, and the interested public. The committee supports scientific progress in astrobiology and planetary science by providing advice to the federal government on the implementation of [decadal survey recommendations](#).

Federal Agencies: DOE, NASA, NSF

- What is the role that federal agencies play in how basic research is done ?

Federal Agencies: DOE, NASA, NSF

- Role that agencies play in how basic research is done
 - Awards – Recognition, Advancement
 - Grants - Faculty, Staff, Students, Postdocs
 - Leadership positions
 - Operate Governmental Labs and Facilities to Manage/Operate majors experiments/telescopes
 - Operate National Computational Centers
 - Fund Major Telescopes/Surveys
 - Fund Major Experiments (LIGO)

Department of Energy (DOE)

- Wikipedia: The DOE is an executive department of the U.S. federal government that oversees national energy policy and energy production, the research and development of nuclear power, the military's nuclear weapons program, nuclear reactor production for the United States Navy, energy-related research, and energy conservation.
- Founded in 1977 – goal to consolidate energy policy
- For some reason, Dark Energy research is funded by DOE
- Dark Matter Particle research is funded and now possibly Dark Matter Astrophysics research.
- Planetary science – funding research in radioisotopes

National Science Foundation

- Strategic Plan: <https://www.nsf.gov/about/performance/strategic-plan/nsf-fy-2026-2030>
- *Wikipedia*: NSF is an independent agency of the United States federal government that supports fundamental research and education **in all the** non-medical fields of science and engineering.
- Established by the [National Science Foundation Act of 1950](#)
- NSF Original Mission: To promote the progress of science; to advance the national **health, prosperity and welfare**; and to secure the national defense
- Over the last 70 years portfolio expanded and charged with **broadening participation** in science, technology, engineering and mathematics.
- Founding premise: that the U.S. Congress embraces the premise that government-funded basic research is essential for the nation's economic health and global competitiveness, and for national defense.

National Science Foundation

Division of Astronomical Sciences <https://www.nsf.gov/mps/ast/about>

- The U.S. National Science Foundation Division of Astronomical Sciences supports leading-edge research in ground-based astronomy and works to ensure the continued scientific excellence of the U.S. astronomical community. The division provides access to world-class research facilities, supports the development of new instrumentation and next-generation facilities and promotes broad understanding of the value and impact of the astronomical sciences.
- Supports:
 - Research Facilities: Ground Based Observatories (Radio, OIR)
 - Centers and Institutes (National AI Research Institute)
 - Workforce Development
 - Spectrum Management: protecting access to the electromagnetic spectrum for research purposes

Planetary Sciences also falls under – Division of Earth Sciences <https://www.nsf.gov/geo/ear> and Division of Atmospheric and Geospace Sciences (AGS) <https://www.nsf.gov/geo/ags>

National Science Foundation

<https://www.nsf.gov/mps/ast/astronomy-astrophysics-advisory-committee>

Astronomy and Astrophysics Advisory Committee (AAAC)

The Astronomy and Astrophysics Advisory Committee (AAAC) is a Federal Advisory Committee that was established by the U.S. Congress under the National Science Foundation Authorization Act of 2002 to provide independent assessments and recommendations regarding the coordination of astronomy and astrophysics programs of the National Science Foundation (NSF), the National Aeronautics and Space Administration (NASA), and the Department of Energy (DOE). As part of its founding charge, the AAAC specifically evaluates the agencies' progress toward the priorities outlined in the National Research Council's decadal surveys in astronomy and astrophysics and transmits its findings and recommendations to the agencies and Congress no later than March 15 of each year.

Still operating in 2026 : [2026 Report](#)

NASA

- National Aeronautics and Space Act of 1958
- “The Congress hereby declares that it is the policy of the United States that activities in space should be devoted to peaceful purposes for the benefit of all mankind.”
- NASA’s formation was directly related to the pressures of national defense (Space Race – Launch of Sputnik)
- The aeronautical and space activities of the United States shall be conducted so as to contribute materially to the following objectives (subset below):
 - (1) The expansion of human knowledge of phenomena in the atmosphere and space;
 - (3) The development and operation of vehicles capable of carrying instruments, equipment, supplies and living organisms through space;
 - (4) The establishment of long-range studies of the potential benefits to be gained from, the opportunities for, and the problems involved in the utilization of aeronautical and space activities for peaceful and scientific purposes.
 - (5) The preservation of the role of the United States as a leader in aeronautical and space science and technology and in the application thereof to the conduct of peaceful activities within and outside the atmosphere.

NASA SMD Advisory Committees : NOW TERMINATED

Astrophysics Advisory Committee. (APAC)

Planetary Science Advisory Committee (PAC)

- <https://science.nasa.gov/researchers/nac/science-advisory-committees/apac/>
- <https://science.nasa.gov/researchers/nac/science-advisory-committees/pac/>
- The APAC supports the advisory needs of the Astrophysics Division (APD), the Science Mission Directorate (SMD) and other mission directorates as required, and NASA Administrator.
- “When it was operating, it was supposed to represent the astronomy community, and also was a forum for NASA to **disseminate** information. So we would hear presentations on how missions in development were going, we would hear about how various postdoc programs were going, we would hear from folks at NASA about the budget, priorities, how they were completing the decadal, etc. ” -- Erika
- UA APAC Past Members: Ilaria Pascucci, Erika Hamden